Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/26/2013
PI Name:	Checka, Neal S.M.		
Project Title:	Automatic Video-based Motion Analysis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	nchecka@vecna.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	(617) 674-8545
Organization Name:	Vecna Technologies, Inc.		
PI Address 1:	6404 Ivy Lane, Suite 500		
PI Address 2:			
PI Web Page:			
City:	Greenbelt	**State:**	MD
Zip Code:	20770-1423	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	01/01/2013	**End Date:**	12/31/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmore, Mihriban	**Contact Phone:**	281-244-1004
Contact Email:	mihriban.whitmore-1@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Gap changes per IRP Rev E (Ed., 3/18/14)		
Key Personnel Changes/Previous PI:	NOTE: PI change in 2015 to C. Mario Christoudias, Ph.D.		
COI Name (Institution):			
Grant/Contract No.:	NNX13CA06C		
Performance Goal No.:			
Performance Goal Text:			
	Understanding task performance and crew behavioral health is crucial to mission success and to the optimal design, development, and operation of next-generation spacecraft. Onboard resources, like a conventional 2D video camera, can capture crew motion and interaction; however, there is a critical need for a software tool that achieves unobtrusive, non-invasive, automatic analysis of crew activity from this footage. The proposed automatic video-based motion analysis software (AVIMA) supports this R&D effort by automatically processing and analyzing complex human motions in conventional 2D video without the use of specialized markers. Unlike many video analytics solutions, AVIMA goes beyond simple blob-based video analysis by tracking the geometric configuration of human body parts like the trunk, head, and limbs. This tracking enables human motion understanding algorithms to model and recognize complex human actions and interactions.		

Task Description:	The resulting system will represent a substantial breakthrough providing benefits to an array of applications in video surveillance, human-computer interaction, human factors engineering, and robotics. POTENTIAL NASA COMMERCIAL APPLICATIONS: Vecna expects the full-scope software system to have immediate and tangible benefit for NASA's Exploration Systems Mission Directorate (ESMD). ESMD focuses on the human element of exploration by conducting research to ensure astronaut explorers are safe, healthy and can perform their work during long-duration space exploration. Task performance and crew behavioral health are key concerns in the design, development, and operation of next generation space vehicles. Operations in confined, isolated, and resource-constrained environments can lead to suboptimal human performance. As such, there is a critical need for Vecna's proposed software tool that automatically processes and analyzes crew motion and interaction from video footage captured by a single conventional 2D video camera. Such a diagnostic tool will enable unobtrusive and non-invasive measurement of task performance and crew behavioral health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed technology is applicable to a wide range of Department of Defense (DoD) and intelligence community areas including force protection, counter-terrorism, human activity monitoring, and surveillance and tracking. We see significant potential for application of this tool to support a range of tactical and strategic systems, including shipboard Navy CIC centers, Army field C3I centers, or USAF theater airborne command posts. A number of programs sponsored by the DoD (FCS, HumanID, CTS, Mind's Eye, Rail Security Pilot) employ video-based monitoring systems and would benefit from the proposed system. Also, Vecna will investigate commercialization opportunities in other sectors, including mobile robotics, interactive displays, and visual surveillance. Initial analysis of these market segments reveal both unaddressed needs as well as vast potential for rapid adoption and growth. As robotic systems become more commonplace in today's society, robust, intelligent interaction between humans and robots is essential. To interact with humans in a lifelike manner requires the robot to infer physical intentions based on visual cues. The proposed technology could potentially revolutionize human-robot interaction. In the surveillance market, automated screening provides an immediate and extant application opportunity for AVIMA. Digital signage and displays provide another venue for applying the AVIMA technology within healthcare.
Task Progress:	New project for FY2013. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)